

Real-Time Game Chat System Production Architecture and

Operating Model Date: 2025-08-10 UTC

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One-Pager Summary

Goal Deliver a real-time, low-latency chat platform for games that supports party, squad, guild, match, global, and direct messages with presence, typing, and moderation, while scaling to millions of concurrent players across regions. The system must be resilient, secure, multi-tenant for multiple titles, and easy to operate with clear SLOs.

Key Outcomes

Sub two hundred millisecond end to end chat delivery in-region and sub fifty millisecond server to client hop. Elastic fanout for large rooms without hot key collapse. Strong correctness with ordered delivery per room, idempotency, and exactly-once effective writes. Safety and trust features including profanity filtering, rate limits, and abuse workflows. Operational excellence with canary releases, automatic rollback, and deep observability.

Primary Assumptions

Concurrent players one million to ten million globally. Peak messages one hundred thousand to one million per second total, regionalized. Message fanout distribution heavy tail for global or guild, majority in small parties or matches. Clients include PC, console, and mobile using secure websockets over TLS or HTTP three for unreliable hints. Consoles may require vendor relay networks and certificate pinning guidelines. Multi-region deployment with data residency options.

Functional Requirements

Send and receive messages in channels party, squad, team, guild, match, lobby, and global, and direct messages. Presence online and offline and away, and match presence. Typing indicators and read receipts optional. Moderation profanity and PII filters, spam throttles, user and word mutes, kicks, bans, and report flows. Persistence recent history per channel with retention policies. Pin and system announcements. Delivery push to online users and queued delivery to offline users with limit and purge rules. Search recent messages within channel scope for players and moderators. Integrations game servers for match or team channels, identity and entitlement, and telemetry. Multi-title tenancy. Per-title and per-environment configuration with quotas and rate limits.

Non-Functional Requirements

Availability read and write APIs ninety nine point nine five percent or better per region. Latency p95 message produce to fanout under one hundred milliseconds in-region. Client receive under two hundred milliseconds end to end in-region. Ordering total order per channel or room, causal in cross channel scenarios. Durability RPO zero for accepted messages with multi AZ. RTO fifteen minutes per region. Scalability linear with partitions and shards. Cost aware with hot cache and tiered storage.

Architecture Overview

High level Regional cells host chat gateways, messaging services, presence, moderation pipeline, cache, and durable stores. A thin global control plane handles auth, config, title and tenant metadata, and routing. Clients connect to the nearest healthy cell through anycast and geo DNS.

Ingestion

and Gateway Edge CDN terminates TLS. API Gateway and Websocket Gateway authenticate via OAuth or platform tokens and enforce per user and per title rate limits and quotas. Session stickiness by user id and room id minimizes cross shard traffic. Backpressure and circuit breakers shed load

gracefully.

Messaging Core Kafka or Pulsar provides ordered, durable streams per channel partition keyed by room id and title id. Producers are chat gateways. Consumers are fanout workers that update Redis lists and publish to per room pub or sub. A transactional outbox in the gateway ensures idempotent publishing. Presence Service Presence state kept in Redis or KeyDB with short TTL heartbeats. Per-player and per-room presence sets. Gateway emits join and leave signals on connect and disconnect. A small state

machine guards ghost sessions.

Moderation Pipeline Inline fast filters for profanity blacklist and repeat detection. Asynchronous deep checks for PII and ML models. Quarantine queues hold suspect messages for moderator review. Muted or banned users are filtered at write and fanout. Abuse reports generate cases in a ticketing system.

Persistence Redis stores hot history windows per room and user inboxes. Durable store is DynamoDB or Scylla or Cassandra with partition keys title id and room id and clustering on time. Cold history moves to object storage in Parquet by day and room for search and compliance.

Data Flow Client sends message to Gateway. Gateway authenticates, runs lightweight filters, and writes to transactional outbox. Outbox writer produces to Kafka partition keyed by room id and title id. Fanout worker consumes, writes to Redis hot history and counters, persists to durable store in batches, and publishes to pub or sub channel. Websocket Gateway pushes to online subscribers. Offline users receive upon reconnect or through mobile push.

Technology Stack Edge and auth Cloud CDN, API Gateway, WAF, OAuth two point one or OpenID Connect, and mTLS in mesh. Streaming Kafka with Schema Registry. Pulsar is an option for geo replication built in. Processing Messaging services in Go or Rust or Java with gRPC internal APIs. Flink optional for analytics taps. Hot cache Redis Cluster for history, presence sets, and counters. Client side hashing and hash tags for room keys. Durable store DynamoDB or Scylla or Cassandra for message history and state. Postgres or Aurora for admin metadata. Search OpenSearch for moderator search across channels and time ranges. Analytics Lake S3 or GCS with Parquet and Iceberg or Delta for long term storage and BI. Platform Kubernetes with autoscaling. Argo CD and Argo Rollouts for delivery. Terraform or Pulumi and Helm for IaC.

Storage Strategy

Hot path Redis lists for recent room history with cap per room. Redis pub or sub for live fanout. Presence in sets and hashes with TTL. Counters for unread and member counts. Durable path DynamoDB or Scylla with per room partitions and sequence to maintain order. Conditional writes enforce monotonic sequences. Batch writers reduce write amplification. Search path OpenSearch indexes recent windows and moderator fields for quick recall. Cold windows are searchable through analytics engines when needed. Cold storage Object storage Parquet partitioned by day, title, environment, region, and room id. Compaction jobs maintain file sizes.

Scaling Approach Partition topics and caches by title id and room id hash. Long lived rooms can be split with sub shard suffixes. Warm shards proactively during events. Presence and history keys use Redis hash tags to keep related keys on the same shard. Fanout workers scale horizontally by partition. Use anycast to route clients to the nearest cell with sticky session to shard. Celebrity rooms use multi tier fanout and partial replication to distribute load.

Data Models

Message title id, environment, room id, message id, user id, sequence, created at, edited at, deleted flag, channel type, body type and body payload json, moderation flags, parent id for threads or replies, custom attributes. Room title id, room id, type party or squad or match or guild or lobby or global, members set, retention policy, slow mode and rate limits, created at. Presence user id, title id, status, last heartbeat at, session id, connection node id. UserModeration user id, title id, mute scopes, ban scopes, expires at, reason. DeliveryReceipt message id, user id, delivered at, read at optional. AuditEvent who, what, target, reason, created at. All records include tenant id and environment, created and updated timestamps, and trace id fields.

Latency Budgets and SLAs

Edge handshake and auth under twenty milliseconds p95. Gateway to Kafka produce under fifteen milliseconds p95. Fanout worker processing under twenty milliseconds p95. Redis append or read under three milliseconds p95. Websocket push under twenty milliseconds p95. End to end

in-region under one hundred milliseconds p95. Availability ninety nine point nine five percent for chat APIs and websocket connections. Data freshness in hot cache immediate upon consume commit.

Key Algorithms and Workflows

Ordering and idempotency Per room sequence numbers with conditional increments. Transactional outbox ensures each message is published once. Consumers dedupe by sequence range. Exactly-once effective writes achieved via idempotency keys and compare and set semantics. Fanout strategy Hybrid fanout on write to Redis for hot rooms and pub or sub. For very large rooms, use tree fanout with tiered relays or room partitions. For small rooms, direct publish suffices. Presence Heartbeats every fifteen seconds with expiry twice the interval. Session fencing ensures reconnects do not duplicate presence. Moderation and safety Inline blacklist and slow mode and spam token buckets protect the hot path. Asynchronous ML checks update the message state. Shadow mutes for suspected bots reduce grieving impact. Backpressure and degradation Drop typing and read receipt signals first. Reduce history window size and sampling for global chat during overload. Enforce per user message budgets with graceful errors.

Security, Reliability, and Compliance

PII minimization store only pseudonymous ids and redact payloads in logs. Encrypt in transit with TLS one point three and at rest with KMS keys, optionally per tenant. RBAC and ABAC for admin tools. Feature gates for risky functions. Multi AZ by default. Regional cells for data residency. Disaster recovery with snapshot and restore for durable store and object storage. WAF and bot protection at the edge.

Multi-Tenancy

Tenant and title id in every key and topic. Soft isolation by namespaces and partitions. Optional hard isolation for premium titles with dedicated cells and VPCs. Per tenant budgets and rate limits. Per title encryption keys and observability slices. CI and CD and Infrastructure as Code Everything as code with Terraform or Pulumi and Helm. Progressive delivery with Argo Rollouts canaries gated on p95 latency, error rate, and Kafka consumer lag. Automatic rollback on SLO budget burn. Schema evolution via registry and contract tests. Online DDL or rolling schema changes for stores. Feature flags and config via control plane. APIs and Integration Points Websocket connect and subscribe to room and presence channels. Publish message, typing, and read receipt events. REST endpoints to create and manage rooms, list history, search, mute, ban, and report. Game server integration API to create match rooms and post system messages. Push integration to mobile platforms when applicable. Telemetry export to analytics lake. Observability and Operational Practices OpenTelemetry for tracing. Prometheus and Grafana dashboards for p95 and p99, Kafka lag, Redis ops, pub or sub throughput, connection counts, and error budgets. Structured logs with tenant and title and room and sequence and trace ids. SLO aware autoscaling for gateways, fanout workers, and brokers. Runbooks for cold cache, shard failover, broker replacement, and regional read only mode. On call rotations and game day drills. Capacity Planning and Cost Estimate Reference busy region with one million concurrent players, ten thousand messages per second sustained and fifty thousand peak. Kafka six to nine brokers with one hundred partitions for hot topics and replication three. Redis twelve shards with less than sixty five percent memory. Durable store cluster sized for write heavy workloads with adaptive capacity. Estimated monthly cost order of magnitude thirty thousand to eighty thousand per busy region depending on retention and search load.

Testing Strategy and Failure Drills

Unit tests for routing, sequencing, moderation gates, and counters. Property tests for ordering and idempotency under retries. Contract tests for gateway and client protocols.

Load tests for hot rooms and celebrity chat. Soak tests to observe memory fragmentation. Chaos drills for broker loss, shard failover, and regional isolation. Disaster recovery drills to verify restore times and

message integrity.

Phased Build Plan

MVP three months single region and title, websocket gateway, party and match chat, Redis hot history, DynamoDB durable store, profanity list, and basic moderation. Minimal search and admin tools. Phase two six to nine months multi region cells, guild and global channels, presence and typing, canary delivery, advanced moderation and abuse workflows, analytics taps, search, and improved history retention. Phase three twelve months hard isolation options per title, ML assisted moderation, tiered fanout for very large rooms, regional data residency, and tighter SLOs for premium titles.

Key Choices, Trade Offs, Risks, and Mitigations Kafka offers replay and ordering at operational cost; mitigated by managed services and focused observability. Redis provides ultra low latency but requires disciplined key design and memory hygiene; mitigated with capped lists, compact payloads, and eviction policies. Exactly once semantics are approximated with outbox and idempotency. Celebrity rooms risk hotspotting; mitigated with tree fanout and partitioning. Compliance with regional rules introduces fragmentation; mitigated with cell architecture and policy enforcement.